25900555 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCPHY201 - ESSENTIAL MATHEMATICS FOR PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **3** marks

1. State Bessel's inequality and explain. [U] / [CO1]

2. The equation of motion of a particle is given by $\mathbf{r}(t) = 2t\ \hat{i} + t^2\ \hat{j} + t^3\ \hat{k}$.Find its velocity and acceleration vectors at $t = 2\,\mathrm{s}$ [A] / [CO1, CO7]

3. Define orthonormality of base vectors. Why is it important in vector spaces? [U] / [CO1]

4. Define an upper triangular matrix and give any one property of it . [U] / [CO2]

5. $\mathbf{A} = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 & -1 \\ 1 & 2 \end{bmatrix}, \text{ and } \mathbf{C} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ [A] / [CO2, CO7]
Calculate D = 2A+B-C.

6. $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.Verify whether the matrix $\mathbf{A}$ is orthogonal. [A] / [CO2, CO7]

7. Write down the expression for the Curl of a vector function in Cylindrical coordinates (r, θ, z). [U] / [CO5]

8. Write down the mathematical expression for the Gradient of a scalar $(\nabla\phi(q_1, q_2, q_3))$ field in general curvilinear coordinate system. [U] / [CO5]

9. $\vec{a} = x^2\hat{i} + yz\hat{j} + z^3\hat{k}$ Find $\nabla(\nabla \cdot \vec{a})$ [A] / [CO4, CO7]

10. State Stoke's theorem? [U] / [CO6, CO7]

11. Write the integral form of divergence of a vector field over a volume. [U] / [CO6]

12. Define the surface integral of a scalar field and a vector field. [K] / [CO6]

[3x10 = 30]

Part B

Short Essay Type Questions

Answer any **8** questions

Each question carries **5** marks

13. Explain the concept of basis vectors. Distinguish between orthogonal and non-orthogonal basis vectors. [U] / [CO1]

14. Show that $[\mathbf{a}, \mathbf{b}, \mathbf{c}] = [\mathbf{b}, \mathbf{c}, \mathbf{a}]$ [U] / [CO1, CO7]

15. Show that the eigen values of a Hermitian matrix are real. [U] / [CO3]

16. Prove that Pauli spin matrices are Hermitian. Where Pauli spin matrices are [A] / [CO2, CO7]

$$\sigma_{\mathbf{x}} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_{\mathbf{y}} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_{\mathbf{z}} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

17. $\mathbf{A} = \begin{bmatrix} 7 & 0 & 6 \\ 0 & -5 & 0 \\ 6 & 0 & -2 \end{bmatrix}$ [U] / [CO3, CO7]

a). What is degenerate eigenvalue.Explain,

b).Show that the matrix **A** has a degenerate eigenvalue

18. The pressure in a fluid is given by $P(x, y) = x^2y$. Find ∇P and explain its significance. [U] / [CO4, CO7]

19. For a rigid body in motion with linear velocity $\vec{V} = \vec{\omega} \times \vec{r}$, where $\vec{\omega} = \omega_x\hat{\mathbf{i}} + \omega_y\hat{\mathbf{j}} + \omega_z\hat{\mathbf{k}}$ and $\vec{r} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\hat{\mathbf{k}}$, show that $\nabla \times \vec{V} = \nabla \times (\vec{\omega} \times \vec{r}) = 2\vec{\omega}$ [A] / [CO4, CO7]

20. A particle moves along a curve defined by $\vec{r}(t) = t^2\hat{i} + 2t\hat{j} + t^3\hat{k}$. Find the velocity and acceleration vectors at t = 1. [A] / [CO4, CO7]

21. $\mathbf{U} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ Find $\mathbf{U}^{-1}$ and verify that $\mathbf{U}^{\dagger}\mathbf{U} = \mathbf{I}$ [A] / [CO3, CO7]

22. A force acts along a straight line according to the law $F(x) = x^3 - 3x(N)$. Determine the net work done as a particle moves from $x = -2 \quad \text{to} \quad x = 2.$ [A] / [CO6]

[5x8 = 40]